REDCap

Knowledge & Experience Portfolio

Research Electronic Data Capture

Demonstrating advanced proficiency in REDCap project design, data collection, reporting, and system administration

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1. Introduction to REDCap

REDCap (Research Electronic Data Capture) is a secure, web-based application developed by Vanderbilt University in 2004, designed specifically to support data capture for research studies and operational data collection. It is hosted and maintained by institutions under a consortium model and is widely used in academic research, public health surveillance, and clinical trials.

My engagement with REDCap began during post-graduate studies and has continued across multiple health projects in the East African context, including malaria surveillance, maternal and child health monitoring, and HIV treatment outcome tracking.

Dimension	Details
Platform	REDCap v14.x (cloud-hosted, institutional server)
Years of Use	2+ years (2024 – present)
Projects Managed	3+ full projects
Primary Role	Project Administrator, Form Designer, Data Manager

WHY *REDCap was selected for these projects because it offers HIPAA/GDPR-compliant data storage, is free to consortium members, supports offline data collection via the REDCap Mobile App, and provides granular user permissions — all of which are critical for multi-site field research.*

2. Project Setup & Configuration

Setting up a REDCap project correctly from the start is foundational. I have led project configuration from scratch, including defining study parameters, selecting the appropriate data collection mode, and configuring event structures for longitudinal studies.

2.1 Project Dashboard

The dashboard provides a real-time overview of project health: total records, instruments, events, and recent user activity. Below is a sample representative dashboard for a malaria surveillance project

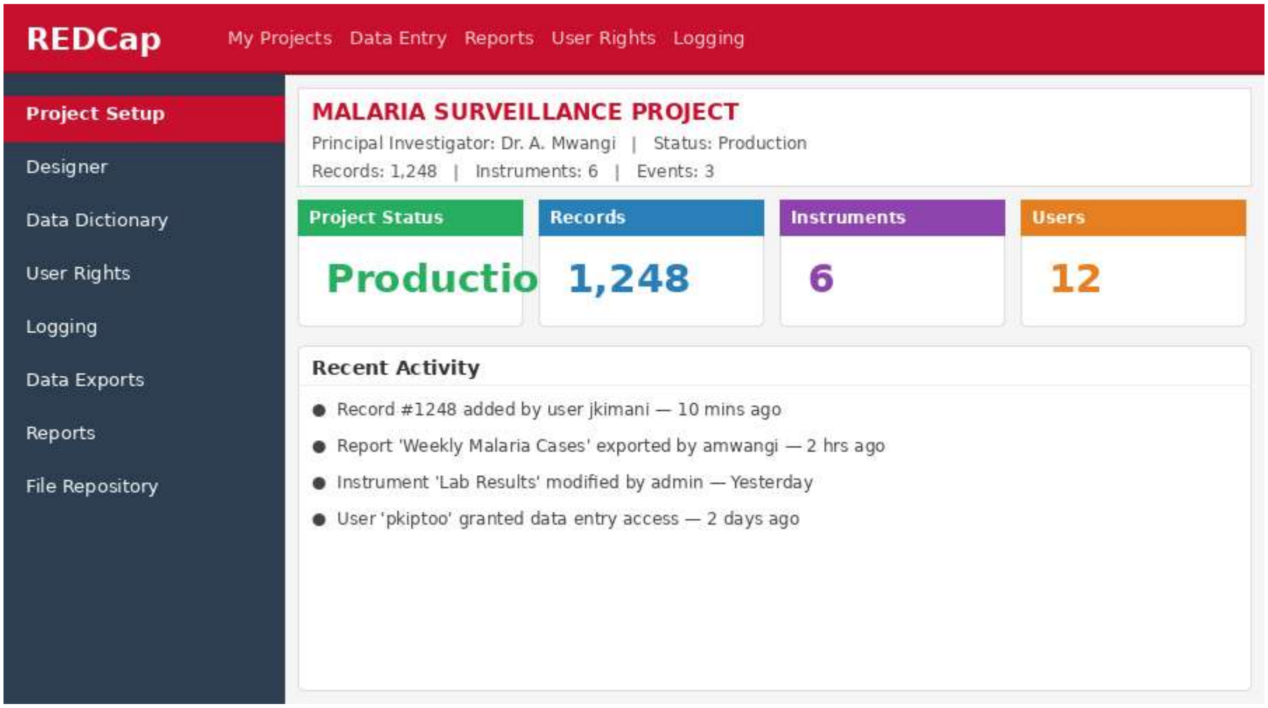


Figure 1: REDCap Project Dashboard — Malaria Surveillance Project

2.2 Key Configuration Decisions

When creating a new project, several architectural decisions must be made upfront:

- Classic vs. Longitudinal: For single-visit studies (e.g., a one-time survey), Classic mode is sufficient. For multi-visit clinical studies, the longitudinal mode with Events and Arms is essential.
- Survey vs. Data Entry: REDCap instruments can be used as internal data entry forms, external patient-facing surveys, or both simultaneously, a hybrid approach I have used for participant self-reporting.
- Project Status: Keeping the project in Development during form building, then moving to Production before going live, which locks the structure and requires formal Change Requests for modifications.
- Randomisation module: For clinical trials, I have configured REDCap's built-in randomisation feature with stratification variables.

Project Type	When to Use	My Experience
Classic (Single Arm)	Cross-sectional surveys, single-visit studies	Used for Community KAP Survey (n=800)
Longitudinal	Multi-visit, cohort, clinical trials	Used for Malaria Follow-Up Study (3 visits)
Survey-only	Public-facing questionnaires	Used for Staff Satisfaction Survey
Hybrid Entry + Survey	Clinic-entered + patient self-report	Used for ANC Monitoring Project

3. Instrument & Form Design

Effective instrument design is the most critical skill in REDCap. Poorly designed forms lead to data quality problems, user frustration, and analytical challenges. I follow a systematic methodology: begin with the Data

Dictionary, design in the Online Designer, validate all field types, then pilot with 10 records before production launch.

3.1 Online Designer

The Online Designer provides an interface for creating fields. I use it for initial construction and rapid iteration. For bulk changes (e.g., standardising variable names across 50 fields), I work directly in the Data Dictionary CSV.

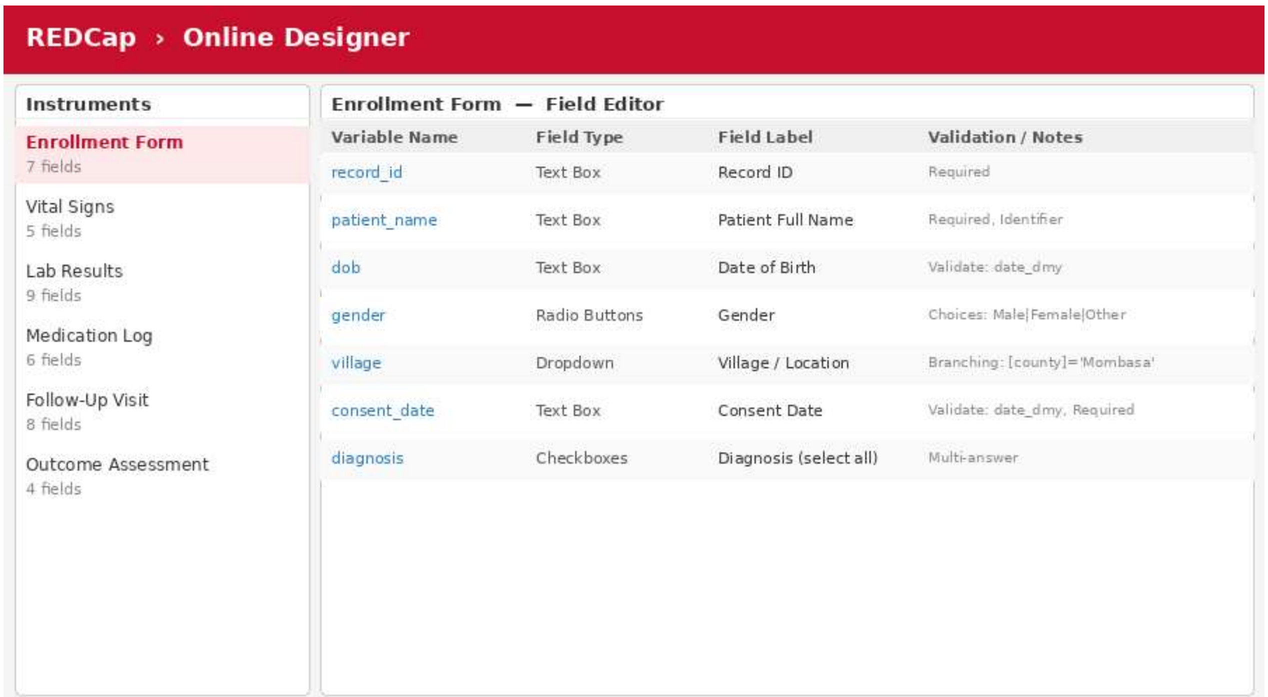


Figure 2: Online Designer — Enrollment Form field editor with variable names, types, and validation rules

3.2 Field Types Used

I have extensive experience with all major REDCap field types, selecting the appropriate type based on the data collection need:

Field Type	Use Case	Validation Applied
Text Box	Names, free text, dates, numbers	Date DMY, integer, email, phone
Notes	Clinical observations, narrative responses	None (free text)
Radio Buttons	Single-select categorical (gender, diagnosis)	Required field + branching
Checkboxes	Multi-select (symptoms, co-morbidities)	Minimum one checked (via branching)
Dropdown	Long lists (county, facility, drug name)	Required; combined with auto-complete
Slider	Pain score, satisfaction rating (0–10)	Min/max bounds
File Upload	Lab reports, consent forms (PDF/image)	Extension whitelist
Calculated Field	BMI, age at enrolment, derived scores	Mathematical expressions
Descriptive	Section headers, instructions, images	N/A (display only)

TIP *I always prefix variable names with instrument abbreviations (e.g., 'enr_dob', 'lab_parasite_count') to make downstream data exports and statistical code self-documenting and to avoid merge conflicts in longitudinal exports.*

3.3 Data Entry Form

Below is an example of a completed data entry form for Lab Results. Note the radio button response for Blood Smear, the structured fields for quantitative results, and the navigation tabs showing instrument completion status across the patient record.

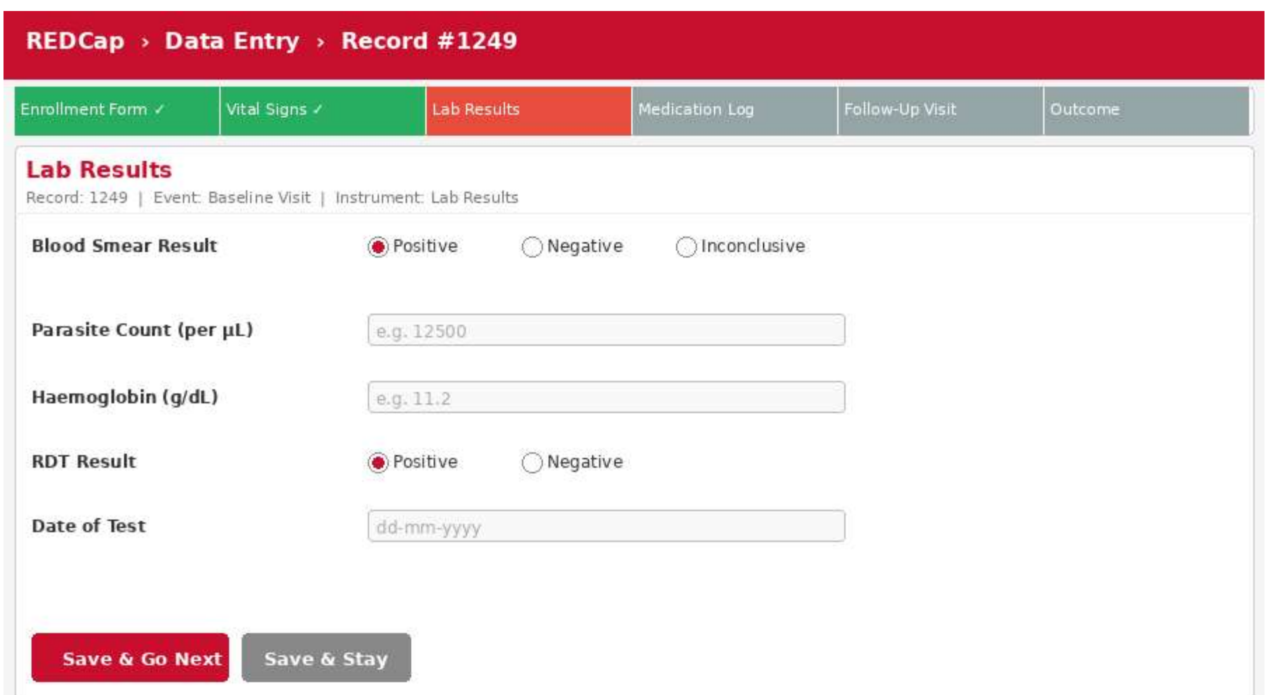


Figure 3: Lab Results data entry form — Record #1249 in the Malaria Surveillance Project

4. Advanced Features

4.1 Branching Logic & Conditional Display

Branching logic in REDCap allows fields to appear or hide based on responses to other fields, reducing form clutter and preventing inapplicable data entry. I have designed complex multi-level branching logic across all my projects.

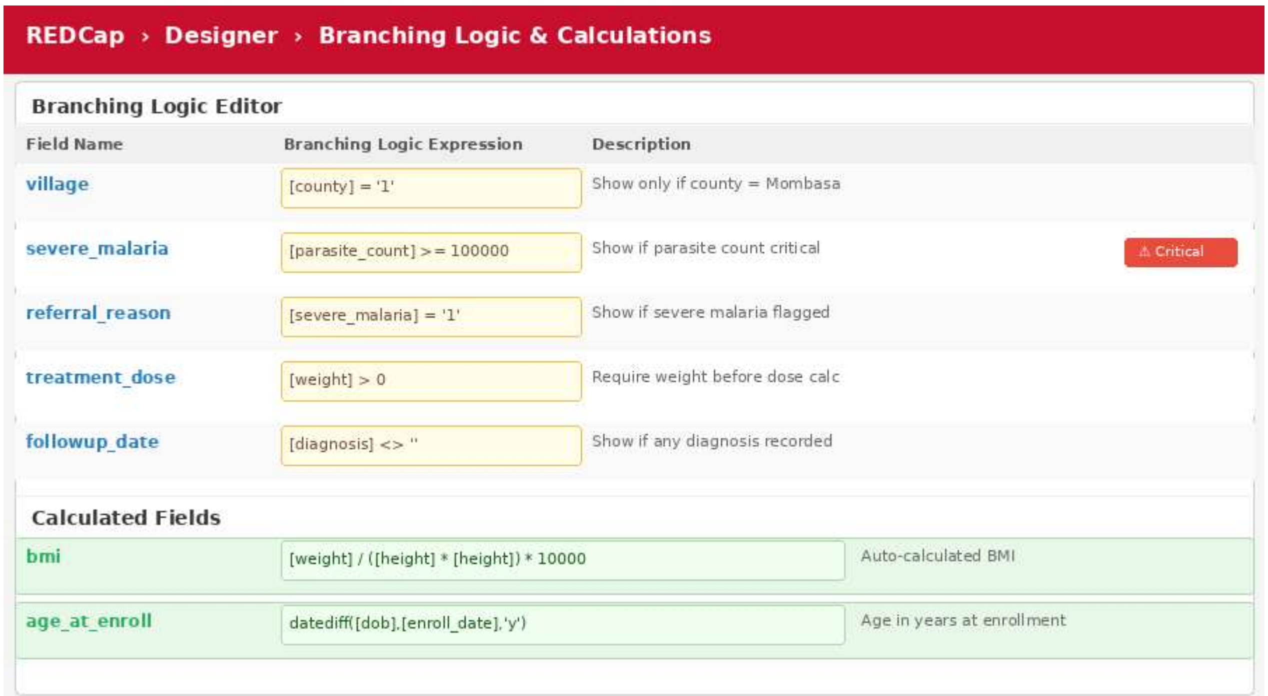


Figure 4: Branching Logic and Calculated Fields editor — showing conditional display rules and computed variables

Example branching logic expressions I have written:

- [county] = '1' — Show village dropdown only for Mombasa County
- [parasite_count] >= 100000 — Flag for severe malaria management pathway
- [consent] = '1' AND [age] >= 18 — Enable adult data entry
- datediff([enroll_date],[today],'d') > 90 — Trigger loss to follow-up alert

BEST PRACTICE *I always test branching logic with edge-case records in Development mode before moving to Production. I document every branching rule in the project's Data Management Plan (DMP) so that future users understand the logic without reverse-engineering the form.*

4.2 Calculated Fields

Calculated fields auto-populate using REDCap's expression language. I have built calculations for:

- Age at enrolment: datediff([dob],[enroll_date],'y',true)
- BMI: [weight_kg] / ([height_m] * [height_m])
- MUAC-for-age z-score (approximated using reference table lookups)

- Days since last visit for follow-up tracking
- Composite risk scores based on multiple clinical variables

4.3 Longitudinal Data Collection — Events & Arms

For cohort studies, I configure REDCap's Longitudinal module with defined Events (e.g., Baseline, Month 3, Month 6, Exit) and Arms (e.g., Intervention, Control). Instruments are mapped to events so that the correct forms appear at each visit, and records track each participant's trajectory over time.

Event	Day	Instruments Mapped	Notes
Screening	Day -7	Screening Form	Assess eligibility
Baseline	Day 0	Enrolment, Vitals, Lab, Medication	Full assessment
Month 1	Day 30 ±3	Vitals, Lab, Follow-Up	Tolerance window ±3 days
Month 3	Day 90 ±7	Vitals, Lab, Follow-Up, Outcome	Primary endpoint
Exit / LTFU	Variable	Outcome Assessment	Early exit or loss to follow-up

4.4 REDCap Mobile App

For field data collection without reliable internet connectivity (common in rural areas), I have configured and deployed the REDCap Mobile App on Android tablets. Key steps include enabling the mobile app in project settings, generating a mobile app API token for each user, and training field workers on offline sync procedures. Data is synced to the server when connectivity is restored.

4.5 Alerts & Notifications

I have configured automated alerts for critical clinical events, including: parasite count exceeding 100,000/µL triggering an SMS to the site clinician, a missed follow-up visit generating an email to the coordinator after 7 days overdue, and record lock notifications sent to the PI upon data verification completion.

5. Reporting & Data Export

REDCap's reporting module allows project teams to monitor data completeness, generate operational summaries, and prepare data for statistical analysis without requiring database query skills.

5.1 Custom Reports

I build custom reports to answer operational and analytical questions, using REDCap's filter logic to subset records by any combination of field values. Reports are shared across the team and can be scheduled for automatic export.

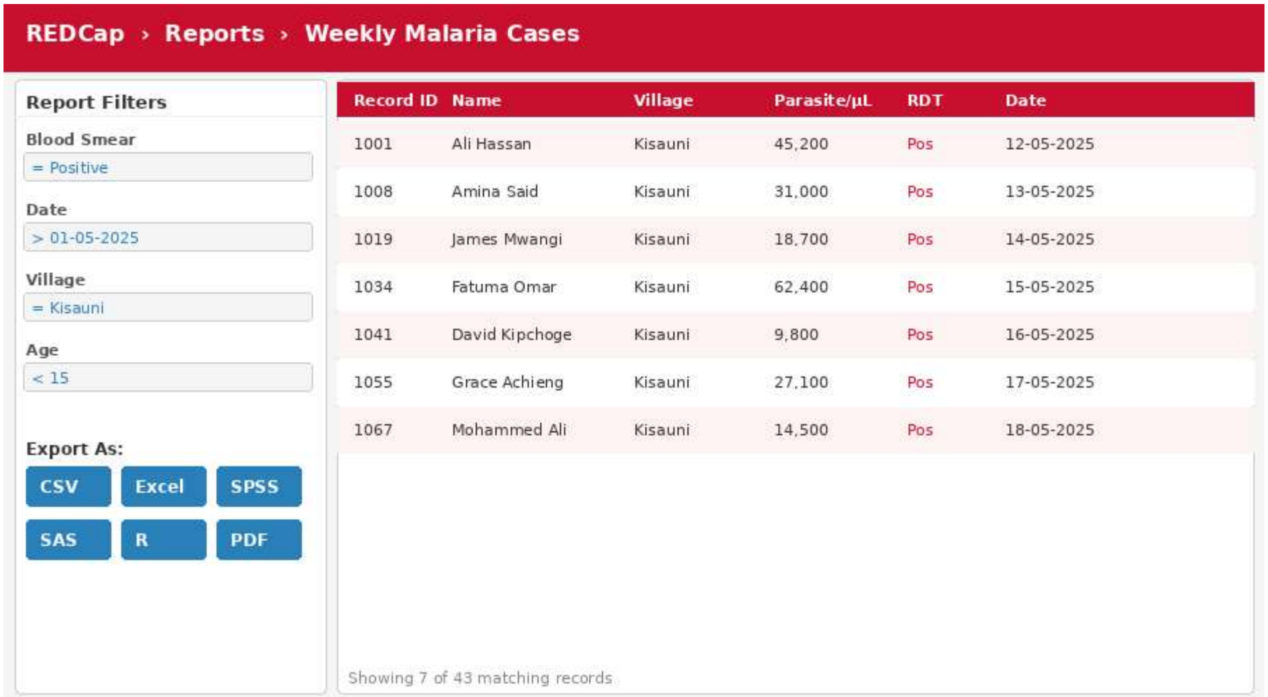


Figure 5: Custom Report — Weekly Malaria Cases filtered by blood smear positive, with export options and data table

5.2 Export Formats

I regularly export data in the following formats for downstream analysis:

Format	Tool Used	Typical Use Case
CSV (Raw)	Excel, Python, R	Cleaning, merging, descriptive statistics
R Script + CSV	R / RStudio	Automated reproducible analysis pipelines
SPSS	IBM SPSS	Shared with biostatisticians who prefer SPSS syntax
SAS	SAS Studio	Regulatory submissions requiring SAS
REDCap JSON	Python (via API)	Automated dashboard refreshes

API *For high-frequency automated exports, one can use REDCap's REST API with Python (PyCap library), scheduling nightly exports via cron to update a Power BI dashboard used by the project PI and funders.*

5.3 Data Quality Checks

Built into my data management workflow are the following quality assurance steps performed within REDCap:

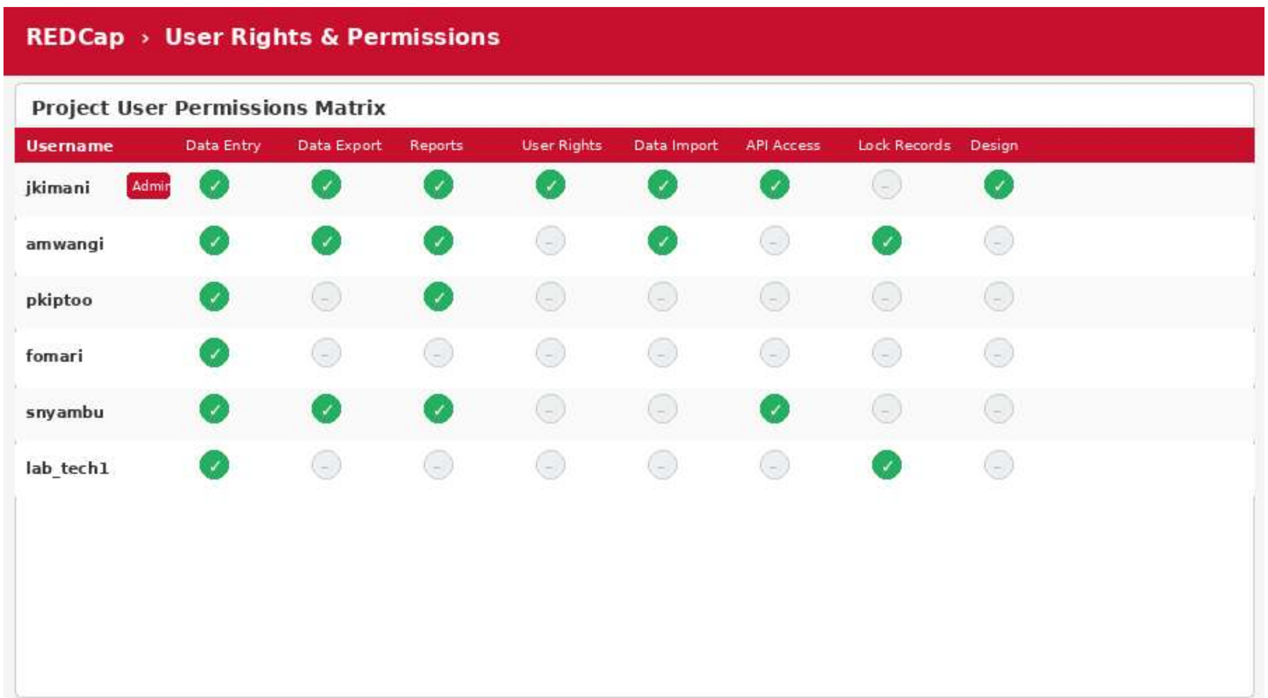
- Data Quality module: Running all 16 built-in rules (missing values, outliers, min/max range violations) weekly
- Field Validation: Setting appropriate min/max for numeric fields (e.g., haemoglobin 3–20 g/dL)
- Required fields: Marking primary outcome fields as required to block save without data
- Double data entry: Enabling REDCap's double-entry verification for primary outcome variables in blinded trials
- Record locking: Locking verified records to prevent retrospective edits

6. User Management & Security

As project administrator, I manage user access with the principle of least privilege; users receive only the permissions required for their specific role. I conduct quarterly access reviews and remove departed staff immediately.

6.1 User Permissions Matrix

REDCap's User Rights module allows granular permission setting per user and per instrument. The screenshot below shows the permissions matrix for the malaria surveillance project team.



REDCap › User Rights & Permissions

Project User Permissions Matrix

Username	Data Entry	Data Export	Reports	User Rights	Data Import	API Access	Lock Records	Design
jkimani Admin	✓	✓	✓	✓	✓	✓	–	✓
amwangi	✓	✓	✓	–	✓	–	✓	–
pkiptoo	✓	–	✓	–	–	–	–	–
fomari	✓	–	–	–	–	–	–	–
snyambu	✓	✓	✓	–	–	✓	–	–
lab_tech1	✓	–	–	–	–	–	✓	–

Figure 6: User Rights & Permissions Matrix — showing role-based access control across the project team

6.2 Role-Based Access Approach

I define role templates before onboarding users, ensuring consistent permissions. Standard roles I implement include:

Role	Permissions Summary	Typical Staff
Project Admin	All permissions, including user rights, design changes, and API	PI, Data Manager
Data Entry Staff	Data entry, no exports, no user management	Field enumerators, clinic staff
Data Reviewer	Read-only access to all forms and reports	Monitors, auditors
Lab Technician	Data entry on Lab forms only (instrument-level)	Lab staff
Statistician	Data export, reports; no entry or user rights	Study biostatistician

SECURITY	*I enforce two-factor authentication for all admin accounts, restrict export access to named investigators, and enable REDCap's logging module so all data access and modifications are auditable — a requirement for GCP-compliant clinical research.*

6.3 Audit Logging

REDCap automatically logs every user action: form saves, field edits, exports, login events, and permission changes. I review the Logging module monthly and before any data lock or regulatory submission to confirm data integrity and chain of custody.

7. REDCap API & Automation

For high-volume or time-sensitive workflows, one can use REDCap's REST API to automate data import, export, and record management. This eliminates manual steps, reduces errors, and allows integration with external systems.

7.1 Common API Operations

Operation	Method	My Use Case
Export Records	POST /api/ (records export)	Nightly export to analytics server
Import Records	POST /api/ (records import)	Bulk import from the lab LIMS system
Export Reports	POST /api/ (report export)	Funder dashboard refresh
Export Metadata	POST /api/ (metadata export)	Data dictionary version control
File Export	POST /api/ (file export)	Retrieve uploaded consent PDFs
Generate Survey Link	POST /api/ (survey link)	Auto-send survey to new participants

Python example (using PyCap) for automated nightly export:

```
import redcap project = redcap.Project(URL, API_TOKEN) records =
project.export_records(format_type="df") records.to_csv("nightly_export.csv",
index=False)
```

8. Challenges Encountered & Solutions

Working with REDCap in resource-limited, multi-site field settings has presented real challenges. Below are possible key challenges that can be encountered, and the possible solutions

Challenge	Context	Solution Implemented
Offline data collection	Rural sites with no internet	Deploy the REDCap Mobile App; establish a daily sync schedule at the district hospital.
Form modification in Production	Needed to add a field after go-live	Use a formal Change Request; add the field without deleting existing data
Duplicate records	Two enumerators entered the same participant	Enable REDCap's built-in secondary ID check; add a deduplication report
Lost API token	Staff member left; token compromised	Revoked token immediately via User Rights; rotate all project API tokens
Branching logic error	Field not showing when expected.	Use REDCap's expression tester; trace variable name typo across 3 instruments.
Large file exports are timing out	12,000 records with file uploads	Switch to chunked API export with date filters; exported in 500-record batches.

9. Conclusion

I am most confident in project administration, instrument design (including complex branching logic and calculated fields), reporting, and API-based automation. I continue to deepen my competency in the randomisation module and advanced survey features such as piping and the Survey Queue.

I am committed to using REDCap according to Good Clinical Practice (GCP) principles, maintaining audit trails, enforcing role-based access, and ensuring data integrity through validated exports and record locking.